**BESTLINK COLLEGE OF THE PHILIPPINES**
IPO Road, Barangay Minuyan Proper City of San Jose Del Monte, Bulacan

# RESEARCH TITLE

# "Integrated HR Management System for Healthway Qualimed Hospital."

In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Information System

**Marycris E. Eliaga**
Project Manager
**Maria Cheska B. Jalotjot**
System Analyst
**Phil Vincent Lope**
Programmer
**Jonha Mhae Azucena**
Business Analyst
**Jade Clarenz Lorenzo**
Document Analyst
**Cedrick Secosana**
Document Analyst

# INTRODUCTION

**Healthway QualiMed Hospital**, located in **Altaraza Town Center, Barangay Tungkong Mangga, San Jose Del Monte, Bulacan**, is a **105-bed general hospital** that provides comprehensive inpatient and outpatient care, emergency services, and various specialty clinics. Part of the **Healthway Medical Network** under **AC Health**, the hospital serves the growing community of Bulacan and nearby towns, relying on a skilled and dedicated workforce of doctors, nurses, allied health professionals, and administrative staff.

Managing this workforce involves many responsibilities, including **hiring, onboarding, attendance tracking, payroll processing, and maintaining employee records**. Currently, many of these tasks are handled manually or through separate tools, which can cause **delays, errors, and increased administrative workload**.

To address these challenges, this study proposes the development of a **Human Resource Management System (HRMS)** tailored to the hospital's needs. The system will automate and centralize HR functions such as **employee information management, time and attendance monitoring, payroll, and performance evaluation**. Implementing the HRMS will improve efficiency, reduce manual work, support better decision-making, and help the hospital maintain its **high standard of patient care**.

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# CHAPTER 1

## 1.1 PROJECT BACKGROUND

In healthcare institutions, **effective human resource management** is essential to support daily operations and maintain quality service. Hospitals handle a large workforce with different roles, making it necessary to organize employee information and HR processes efficiently.

Many organizations still rely on **manual Human resource processes or separate tools**, which can lead to data duplication, delayed reporting, and increased administrative workload. These challenges highlight the need for a more **integrated and reliable system** to manage employee-related information.

The development of a **Human Resource Management System (HRMS)** can help address these issues by centralizing HR data and automating important processes such as employee onboarding, scheduling, performance monitoring, and payroll management. By implementing this system, organizations can improve accuracy, reduce manual work, and support better decision-making in human resource management.

### 1.1.1 Problem / Opportunity Description

**Healthway QualiMed Hospital** in **San Jose Del Monte** currently handles most HR tasks manually, including **recruitment, onboarding, attendance tracking, employee records, payroll, employee portal management, and employee separation**. These paper-based or spreadsheet methods are inefficient, prone to errors, and create extra work for HR staff. They also make it harder to maintain accurate records and ensure compliance with regulations.

As a result, the hospital experiences the following problems:

- **HR processes are not connected** – Recruitment, onboarding, payroll, employee portal, and other HR tasks are handled separately, causing inefficiencies.
- **Manual HR work is slow and repetitive** – Doing tasks by hand increases workload and delays processes.
- **Data errors and compliance risks** – Employee records are scattered or outdated, reducing reporting reliability and creating potential compliance issues.
- **Employee information is scattered** – HR staff lack a single source of truth for employee profiles, schedules, and records, making planning and decision-making difficult.
- **Suboptimal employee experience** – Staff must navigate slow and cumbersome processes to access information, limiting engagement and effective use of the employee portal.
- **Challenges in supporting hospital growth** – Manual processes make recruitment, training, and retention difficult as the hospital expands.
- **Limited scalability and adaptability** – Current HR processes cannot easily evolve with the hospital's growing operational requirements.

### 1.1.2 Benefits

The implementation of the proposed **Human Resource Management System (HRMS)** will provide significant benefits to **Healthway QualiMed Hospital, San Jose Del Monte**, including:

1. **Optimized Recruitment and Candidate Management** – Automating online job postings, applicant tracking, and initial screening reduces time-to-hire for critical clinical and support positions, ensuring the hospital maintains adequate staffing for uninterrupted patient care.
2. **Improved Selection Accuracy and Decision Support** – Structured interview scheduling, candidate evaluation scoring, and selection analytics improve hiring

quality and align new hires with departmental needs, minimizing mismatches in critical hospital roles.

3. **Faster Onboarding and Compliance Readiness** – Digitized new employee orientation, electronic document management, and automated training assignments accelerate staff integration, ensuring compliance with hospital credentialing and regulatory requirements.
4. **Reliable Timekeeping and Workforce Utilization** – Automated clock-in/out, leave monitoring, and attendance reporting enhance shift management, reduce payroll errors, and ensure proper staffing coverage in patient care areas.
5. **Centralized Employee Management and Performance Oversight** – Consolidated personnel records, performance tracking, and organizational structure management provide HR and hospital leadership with actionable data for workforce planning, employee development, and retention strategies.

### 1.1.3 Goals

The primary goals of this HRMS project are to:

- **Integrate core HR functions** – Automate recruitment, selection, onboarding, time and attendance monitoring, employee records management, and termination processes.
- **Streamline HR operations** – Reduce manual and repetitive tasks to improve efficiency and save administrative time.
- **Ensure accuracy and compliance** – Maintain reliable data, generate automated reports, and align with hospital policies and labor regulations.
- **Centralize employee information** – Provide a unified database to support managerial decision-making and workforce planning.
- **Enhance employee experience** – Offer self-service features and simplified administrative workflows.

- **Support long-term hospital growth** – Improve recruitment, retention, and workforce development.
- **Enable scalability and adaptability** – Build a system capable of evolving with future operational requirements.

## 1.1 PROJECT SCOPE

The scope of this project covers the design, development, and implementation of a comprehensive **Human Resource Management System (HRMS)** for **Healthway QualiMed Hospital, San Jose Del Monte**. The system is intended to automate and centralize key human resource processes, enhancing efficiency, improving data accuracy, and supporting workforce management while ensuring compliance with institutional and regulatory standards.

### 1.2.1 Scope and Objectives

**Scope:** The scope of this project covers the development and implementation of a complete Human Resource Management System (HRMS) for Healthway QualiMed Hospital, San Jose Del Monte.

| Project Scope | Description |
|---|---|
| Job Posting Management | Allows HR personnel to create, publish, update, and manage job postings to attract potential applicants. |
| Recruitment Management | Supports applicant tracking and the initial screening of candidates to streamline the recruitment process. |

| | |
|---|---|
| Selection Management | Facilitates interview and assessment scheduling, sends email invitations to candidates, manages candidate evaluations, and provides decision support for selecting qualified applicants. |
| Contract Generation | Generates employment contracts for selected candidates, sends them automatically to the employee's registered email, and creates employee access credentials for the self-service portal. |
| Onboarding Management | Manages onboarding tasks and the submission of required documents for newly hired employees. |
| Employee Management | Maintains detailed employee records including personal information, contact details, emergency contacts, government identification numbers, employment details, schedules, salary information, and required employee documents. |
| Employee Self-Service Portal | Provides employees with secure access to a personalized dashboard where they can view and update their profile, check leave balances, submit leave and reimbursement requests, access payroll information and payslips, view work schedules, enroll in trainings, monitor onboarding tasks, and manage account settings. |

| | |
|---|---|
| Attendance Management (HR/Admin) | Allows HR to monitor overall employee attendance, including present, late, absent, half-day, and leave status. HR can search employees, filter by date, view shift schedules, approve/decline leave requests, and manually adjust attendance records. |
| Time and Attendance (TimeKey / Employee Clock-In) | Allows employees to clock in and out, track hours worked, view workdays, and monitor attendance status (on time, late, etc.). |
| Payroll and Compensation | Processes payroll for multiple employees, calculates regular, overtime, night, and holiday pay based on hours worked, deducts applicable contributions (SSS, PhilHealth, Pag-IBIG, taxes), generates payroll results and payslips, and supports batch processing by cutoff periods. Includes configuration of compensation components such as employee benefits, allowances, taxes, holiday pay, differentials, and reimbursements. HR can set values as fixed amounts or percentages and define whether they are taxable. |
| Benefits and Reimbursements | Manages employee benefit enrollment and reimbursement requests with corresponding approval workflows. |
| Termination Management | Handles employee offboarding procedures, exit interviews, and calculation of final settlements during employee separation. |
| Reporting (HR Dashboard) | Provides HR with a dashboard summarizing key metrics such as total employees, active |

| | |
|---|---|
| | positions, total candidates, pending interviews, pending leave requests, training programs, recent hires, payroll overview, employee logins, and time & attendance summaries. |

*Scope Table No. 1*

**Objective:** The main objective of this project is to deliver a **fully functional Human Resource Management System (HRMS)** that supports Healthway QualiMed Hospital's operational needs, streamlines HR processes, ensures compliance with labor and regulatory requirements, and enables strategic HR management through automation and centralized data management.

| Project Deliverables | Description |
|---|---|
| HRMS Software Application | Complete web-based system including all modules: Job Posting, Recruitment, Selection, Contract Generation, Onboarding (document submission), Employee Management, Employee Self-Service Portal, Attendance Management (HR/Admin), Time and Attendance (TimeKey), Payroll & Compensation, Benefits & Reimbursements, Termination Management, and Reporting (HR Dashboard). All modules include user interfaces, workflows, and automation features. |
| Database Design & Implementation | Secure, relational database designed to store employee records, HR data, payroll |

| | |
|---|---|
| | information, attendance logs, training records, and system audit trails with backup and recovery capabilities. |
| User Documentation | Comprehensive manuals, help guides, and training materials for HR staff, administrators, and employees covering system navigation, portal usage, leave and reimbursement requests, attendance tracking, and reporting functions. |
| Technical Documentation | Detailed system architecture, API specifications, module descriptions, integration guidelines, security configurations, and maintenance procedures. |
| Testing Reports | Results from unit testing, integration testing, user acceptance testing, and validation of all HRMS modules to ensure functionality, data integrity, and regulatory compliance. |
| Deployment Package | Installation scripts, configuration files, and deployment procedures for full system implementation on hospital servers or cloud infrastructure. |
| Training Sessions | Conduct initial training for HR staff, administrators, and end-users covering all HRMS modules, including Employee Self-Service Portal usage, Payroll operations, Attendance tracking, compliance-related trainings, and handling automated workflows. |

*Objective Table no. 1*

**Approach and Methodology**

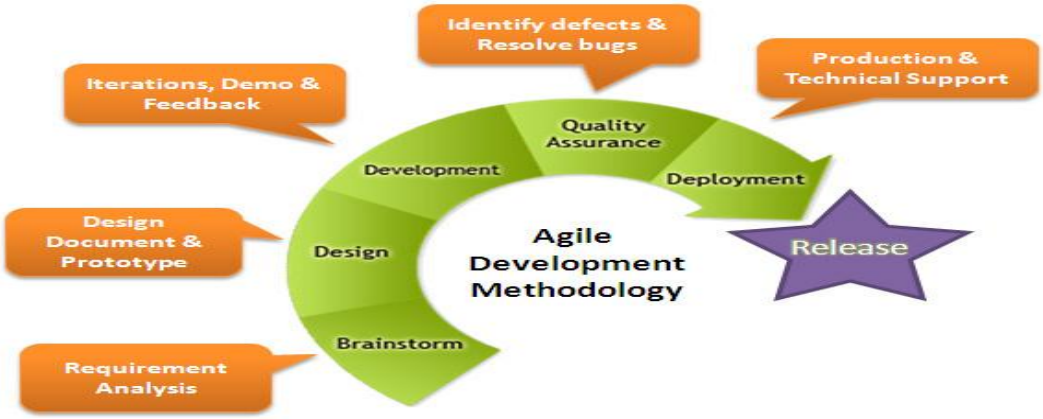



*Figure No. 1 Approach and Methodology*

- **Brainstorm**
  Is a group or individual creativity technique by which efforts are made to find a conclusion for a specific problem by gathering a list of ideas spontaneously contributed by its member sis critical to the success or failure of a systems or a software project. The requirements should be documented, actionable, measurable, testable, traceable, related to identified business needs or opportunities, and defined to a level of detail sufficient for system design.
- **Design**
  In this stage the developers create a prototype of a system that will be use; the developers will carefully design the system and make sure the process is followed.
- **Development**
  In this stage the developers will test the proposed system to make sure it completes all the requirements. Then will analyze the feedback to do what can be done to improve the system

- **Quality Assurance**
  In this stage the developers will identify defects and will resolve bugs or error that encounter before deployment. Once the bug is identified the developers will then fix the problem and will develop a solution that will make bugs and other errors disappear.
- **Deployment**

In this stage the system is ready to use and implement and after the implementation the developers will give updates and technical support.

### 1.3.2 PROJECT TIMELINE

The following timeline outlines the planned activities, start dates, duration, and status for the development and implementation of the **Human Resource Management System (HRMS)** for **Healthway QualiMed Hospital, San Jose Del Monte**.

| Activity | Plan Start Hours | Plan Duration Hours | Actual Start Day | Actual End Day | Status |
|---|---|---|---|---|---|
| **Alumni System Documentation** | | | | | |
| **Chapter 1**<br>**Chapter 2**<br>**Chapter 3** | | | | | |
| Data Gathering | Jan 10 (Fri) | 2 weeks | Jan 10 (Fri) | Jan 15 (Thu) | Completed |
| Planning | Jan 11 (Sat) | 8 | Jan 11 (Sat) | Jan 11 (Sat) | Completed |
| Monitoring | Jan 12 (Sun) | - | Jan 12 – Mar 18 | Mar 18 (Mon) | Completed |
| Execute | Jan 20 – Mar 18 | - | Jan 20 (Tue) | Mar 18 (Mon) | Completed |
| Design Effective | | | | | |
| Job Posting Management | Jan 26 (Sun) | 1 day | Jan 26 (Sun) | Jan 26 (Sun) | Completed |
| Recruitment Management | Jan 27 (Mon) | 1 day | Jan 27 (Mon) | Jan 27 (Mon) | Completed |
| Selection Management | Jan 28 (Tue) | 1 day | Jan 28 (Tue) | Jan 29 (Wed) | Completed |
| Contract Generation | Jan 30 (Thu) | 1 day | Jan 30 (Thu) | Jan 30 (Thu) | Completed |

| | | | | | |
|---|---|---|---|---|---|
| Onboarding Management | Jan 31 (Fri) | 1 day | Jan 31 (Fri) | Jan 31 (Fri) | Completed |
| Employee Management | Feb 1-2 (Sat-Sun) | 2 days | Feb 1 (Sat) | Feb 2 (Sun) | Completed |
| Employee Self-Service Portal | Feb 3 (Mon) | 1 day | Feb 3 (Mon) | Feb 3 (Mon) | Completed |
| Attendance Management (HR/Admin) | Feb 4 (Tue) | 1 day | Feb 4 (Tue) | Feb 4 (Tue) | Completed |
| Time and Attendance (TimeKey / Employee Clock-In) | Feb 5 (Wed) | 1 day | Feb 5 (Wed) | Feb 5 (Wed) | Completed |
| Payroll and Compensation | Feb 6-7 (Thu-Fri) | 1 day | Feb 6 (Thu) | Feb 7 (Fri) | Completed |
| Benefits and Reimbursements | Feb 8 (Sat) | 1 day | Feb 8 (Sat) | Feb 8 (Sat) | Completed |
| Termination Management | Feb 9 (Sun) | 1 day | Feb 9 (Sun) | Feb 9 (Sun) | Completed |
| Reporting (HR Dashboard) | Feb 10 (Mon) | 1 day | Feb 10 (Mon) | Feb 10 (Mon) | Completed |
| Final Review & Integration | Mar 11-18 | - | Mar 11 (Mon) | Mar 18 (Mon) | Completed |

*Project Timeline Table No. 1*

### 1.3.2 ISSUES AND POLICY IMPLICATIONS

The implementation of the Human Resource Management System (HRMS) at **Healthway QualiMed Hospital, San Jose Del Monte** involves multiple modules that streamline HR processes. Several issues and policy considerations must be addressed to ensure effective adoption and smooth operations:

1. **Data Privacy & Security** – The HRMS manages sensitive employee information, including personal details, government IDs, salary information, and payroll data.

Strict policies and secure storage are essential to prevent unauthorized access or data breaches. Role-based access controls in the employee self-service portal must also be enforced.

2. **Change Management** – Employees and HR staff must adapt to the new system for job posting, recruitment, selection, document submission, attendance tracking, and payroll management. Resistance to change could reduce system adoption and the efficiency of automated workflows.
3. **Regulatory Compliance** – The HRMS must comply with labor laws, government contribution requirements (SSS, PhilHealth, Pag-IBIG), payroll regulations, and internal HR policies. Modules like Payroll & Compensation and Termination Management must accurately reflect legal obligations. Additionally, the system should support tracking of **mandatory trainings, drills, and certifications required by labor regulations**, including occupational safety, medical and fire drills, and other workplace compliance programs.
4. **System Integration** – The HRMS needs to integrate with existing hospital systems, such as attendance tracking devices (TimeKey), payroll software, and reporting dashboards, to maintain seamless HR operations.
5. **Resource Allocation** – Successful implementation requires adequate resources, including technical support, employee training for portal usage, and HR staff involvement in modules like Recruitment, Selection, and Employee Management.
6. **Operational Disruption** – During the transition, manual and automated processes may run concurrently. Careful scheduling and monitoring are required to prevent delays in recruitment, onboarding, attendance reporting, payroll processing, and other HR operations.

**Policy Implications**

- Establish **data access and security policies** covering all modules, especially payroll, employee records, and the self-service portal, with role-based permissions.

- Update HR policies to incorporate **automated workflows**, such as document submission, leave requests, training tracking, and approval processes.
- Define policies for **system usage, change communication, and escalation procedures** to ensure consistent operation across modules including Recruitment, Selection, Attendance, Payroll, and Reporting.
- Provide guidance on **resource allocation and training**, ensuring HR staff and employees can efficiently use the portal, manage contracts, track attendance, complete mandated trainings and drills, and handle payroll-related tasks.

### 1.3.4 RISK MANAGEMENT PLAN

The following table identifies potential risks in the implementation of the **Human Resource Management System (HRMS)** at **Healthway QualiMed Hospital, San Jose Del Monte**, along with their probability, impact, and proposed mitigation actions:

| Risk Factor | Probability (H-M-L) | Impact (H-M-L) | Risk Management Action |
|---|---|---|---|
| Data breaches / unauthorized access | M | H | Implement role-based access across all modules (Employee Portal, Payroll, Employee Records), encrypt sensitive data, conduct regular security audits, and enforce strong password policies. |
| System downtime / technical failures | M | H | Schedule regular backups, use reliable servers and hosting, implement monitoring, and provide IT support for quick recovery in modules like Attendance, Payroll, and Reporting. |
| Errors in HR processes | L | H | Test all modules extensively (Job Posting, Recruitment, Selection, Payroll, Benefits, |

| | | | |
|---|---|---|---|
| | | | Termination), validate formulas, and perform parallel runs with manual verification before full deployment. |
| Integration issues with existing systems | M | M | Conduct system integration testing, plan phased integration with existing hospital systems (TimeKey, payroll software, attendance devices), and provide clear API documentation. |
| Loss or corruption of employee data during migration | L | M | Perform multiple backups before migration, test migration in a sandbox environment, and verify data integrity post-migration for Employee Records, Payroll, and Attendance modules. |
| Insufficient system scalability | M | M | Design the system modularly, plan for future growth, and ensure database and server resources can be expanded to accommodate increasing users and data from all modules. |
| Regulatory compliance violations | L | H | Implement automated compliance checks in Payroll, Benefits, and Termination modules, maintain audit trails, and periodically review labor laws, government contributions (SSS, PhilHealth, Pag-IBIG), and hospital policies. Include monitoring of mandatory trainings and labor-required drills. |

| | | | |
|---|---|---|---|
| Unauthorized modifications or errors by users | M | M | Implement role-based permissions, log all user activities in Employee Portal and HR modules, and enforce approval workflows for critical data changes such as payroll updates or contract generation. |
| Cybersecurity attacks (ransomware, phishing) | M | H | Conduct staff cybersecurity training, deploy anti-malware and firewall protections, and maintain a response plan covering all HRMS modules. |

*Risk Management Plan Table no. 1*

### 1.3.2 SERVICE TRANSITION

The **Service Transition** phase focuses on moving the Human Resource Management System (HRMS) from development to full operational use at **Healthway QualiMed Hospital, San Jose Del Monte**, ensuring that all modules function smoothly and support hospital staff without disruption. Key activities during this phase include:

1. **User Training** – HR staff, department heads, and employees will receive comprehensive training on all HRMS modules, including **Job Posting, Recruitment, Selection, Contract Generation, Onboarding (document submission), Employee Management, Employee Self-Service Portal, Attendance Management, Time and Attendance (TimeKey), Payroll & Compensation, Benefits & Reimbursements, Termination Management, and Reporting**. Training covers system navigation, workflows, approvals, payroll calculations, leave/reimbursement processing, and compliance tracking.
2. **Data Migration** – Existing employee records, attendance logs, payroll data, training and drill records, and other HR documents will be securely transferred to the new system, ensuring data accuracy, completeness, and integrity.

3. **System Testing & Validation** – Final testing will verify that all features and modules operate correctly in the hospital environment, including **approvals, automated notifications, payroll computations, contract generation, and reporting dashboards**. Any discrepancies will be resolved before full deployment.
4. **Technical Support** – IT support will be available throughout the transition to address user issues, resolve errors, assist with portal access, and provide guidance for all modules.
5. **Feedback & Adjustments** – Post-deployment, feedback from HR staff and employees will be collected to fix bugs, optimize workflows, improve usability, and enhance system performance. Continuous monitoring ensures compliance with hospital policies and labor regulations.

## 1.4 TECHNICAL FEATURES

| Category | Description |
|---|---|
| **Hardware Specification** | **Desktop Computers / Workstations:**<br>• Minimum: Intel i3 / AMD Ryzen 3, 8GB RAM, 256GB SSD, 21” monitor, Ethernet/Wi-Fi, Windows 10 or equivalent OS<br>• Recommended: Intel i5 / AMD Ryzen 5, 16GB RAM, 512GB SSD, 24” monitor, Ethernet/Wi-Fi, Windows 11 Pro or equivalent OS<br>**Biometric Attendance Devices (Optional):** Fingerprint scanner or facial recognition for accurate attendance tracking<br>**Network Server:** 8-core CPU, 16GB RAM, 1TB storage, RAID configuration for backup, secure network connection<br>**Backup Storage:** External drives or NAS with automated daily backups for disaster recovery<br>**Internet Connectivity:** Stable high-speed connection for cloud modules or remote access |

| | |
|---|---|
| **Software** | **Web-based HRMS application** supporting all modules (*Recruitment, Selection, Onboarding, Time & Attendance, Employee Management, Termination*)<br>**Database Software:** MySQL, PostgreSQL, or equivalent<br>**Security Software:** Anti-virus, encryption, firewall, and access control<br>**Reporting Tools:** Dashboards, analytics, and HR reports.<br>**Optional:** Cloud integration for remote access and system updates |

*Technical Features Table no. 1*

## 1.5 PROJECT ORGANIZATION AND STAFFING

| ROLE | NAMES & CONTACT INFORMATION | RESPONSIBILITIES |
|---|---|---|
| Scrum Master | Maria Cheska Jalotjot<br>jalotjot.cheska@icloud.com<br>0994-181-3832 | • Report to and receive direction from sponsors.<br>• Manage, review, and prioritize project work plans.<br>• Provide status reports.<br>• Manage project team.<br>• Recommend changes, escalate issues, and mitigate risks. |
| Development Team | Marycris Eliaga<br>eliaga.marycris@gmail.com<br>09994457694 | • Participate in project activities, including *planning*, *implementation of deliverables*, and *quality control*. |
| Stakeholders | Healthway Medical Network<br>care@healthway.com.ph<br>0917-466-2273 | • Review and approve the final system. |

*Project Organization and Staffing Table no. 1*

## 1.6 PROJECT BUDGET

| Budget Item | Description | Budgeted Cost |
|---|---|---|
| **System Development** | Costs for designing, coding, configuring, and testing all HRMS modules, including **Job Posting, Recruitment, Selection, Contract Generation, Onboarding (document submission), Employee Management, Employee Self-Service Portal, Attendance Management, Time and Attendance (TimeKey), Payroll & Compensation, Benefits & Reimbursements, Termination Management, and Reporting**. Includes labor costs for the development and HR implementation team, as well as integration, automated workflows, compliance tracking, and system documentation. | 250,000 |
| **Hardware Installation** | Procurement and setup of servers, computers, networking devices, biometric scanners for | 120,000 |

| | | |
|---|---|---|
| | time & attendance, and other necessary equipment to support system deployment and employee access. | |
| **Software Installation** | Costs for installing required software, licenses, database systems, HRMS tools, cybersecurity measures, and integration with existing hospital systems. | 80,000 |
| **Total Costs** | | 450,000 |

*Project Budget Table no. 1*

# CHAPTER 2 REVIEW OF RELATED STUDIES AND PROJECTS/SYSTEMS

## *2.1 LOCAL*

Provide at least three (3) LOCAL related studies and projects/systems.

**Local Related Study 1.**

**"Employee Performance Data Management through HRIS at Ilocos Sur Polytechnic State College (Philippines)."**

A descriptive and developmental research that explores the use of a Human Resource Information System (HRIS) for managing employee performance data, showing how HRIS-generated reports aid decision-making and planning. (View of Employee

Performance Data Management and Analysis of Ilocos Sur Polytechnic State College Through Human Resource Information System Application, n.d.)

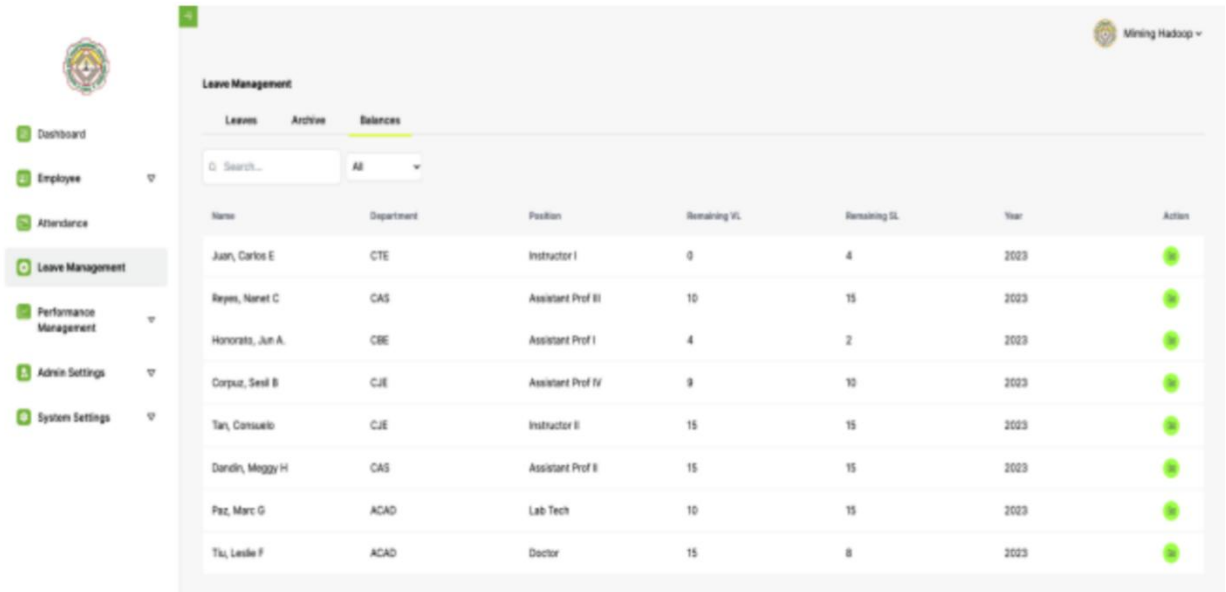



*Figure 1.*

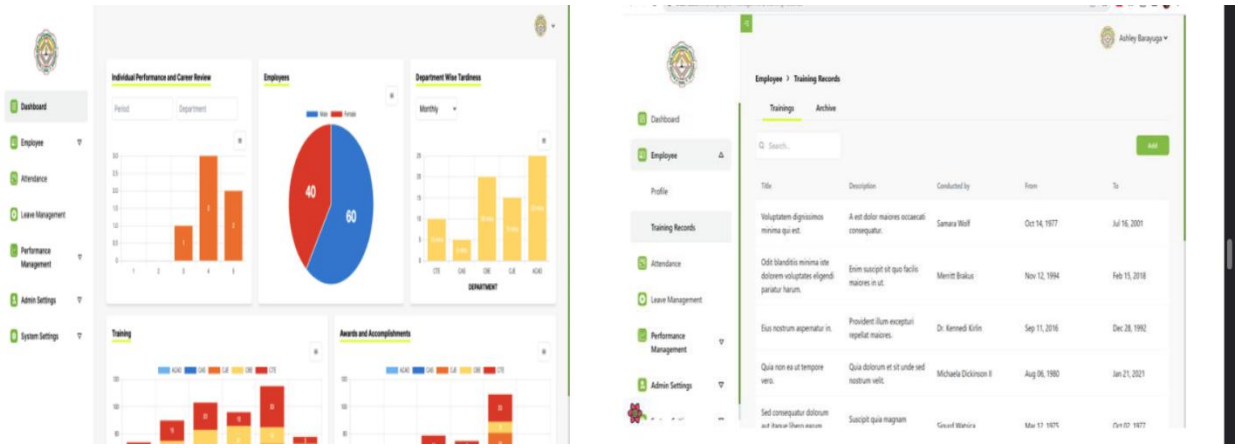



*Figure 2.*

**Local Related Study 2.**

**"Implementation of Electronic Human Resource Management (E- HRM): Its Effectiveness and Challenges."**

A recent study (2024) examining the benefits and challenges of e-HRM system adoption in banks in the Philippines, including system tools and impacts on HR operations.

Bonifacio, R. V., & Dm, E. M. M. P. (2025). Implementation of Electronic Human Resource Management (E- HRM): Its effectiveness and challenges. *International Journal of Science and Management Studies (IJSMS)*, 164–182. https://doi.org/10.51386/25815946/ijsms-v8i2p116

BONIFACIO, RALPH V. "Implementation of Electronic Human Resource Management (E-HRM): Its Effectiveness and Challenges". A Doctor of Management Major in Human Resource Management Dissertation, Guimaras State University, Buenavista, Guimaras, Philippines, February, 2025.

Accepted and approved in partial fulfillment of the requirements for the degree, Doctor of Management Major in Human Resource Management.

Comprehensive Examination passed on September 28-29, 2024 .
Date of Final Oral Examination: March, 02, 2025.

**ERLY M. MARTIR, Ph.D., DM**
Dean, Graduate School

**ABSTRACT**

*This study examined the implementation, effectiveness, and challenges of electronic Human Resource Management (e-HRM) in Higher Education Institutions (HEIs) in Aklan, Philippines, during the 2024-2025 academic year. Utilizing a descriptive-correlational design, the research involved teaching and non-teaching personnel from selected HEIs, with convenience sampling employed to gather data. A researcher-made survey instrument was used to assess e-HRM implementation, effectiveness, and challenges. Data analysis included descriptive statistics, the Mann-Whitney U test, the Kruskal-Wallis test, and Spearman's rho correlation, with hypotheses tested at a 0.05 significance level. Findings revealed that e-HRM implementation was rated as "Very High" across operational, relational, and transformational dimensions, indicating strong integration within HEIs. The system effectively enhanced HR processes, communication, and strategic alignment, with teaching personnel, public HEI employees, and supervisory staff reporting slightly higher perceptions of effectiveness. Despite its strengths, e-HRM posed significant challenges, particularly in navigation difficulties, technical support limitations, and communication barriers, with public HEIs facing greater obstacles. Statistical analyses showed no significant differences in implementation perceptions based on demographic factors, while effectiveness varied by job position. Additionally, a weak positive correlation was found between implementation and effectiveness, but challenges remained independent of these variables. The study concluded that while e-HRM significantly improved HR operations, persistent challenges necessitated targeted interventions. Recommendations included enhanced user training, technical support, policy development, and infrastructure improvements to optimize e-HRM utilization. Future research could explore AI-driven HR systems and public HEI-specific challenges to further refine institutional HR*

*Figure 3.*

**Local Related Study 2.**

**"Strategic Human Resource Management Framework for State Universities and Colleges in the Philippines."**

This qualitative study discusses HR challenges in recruitment, selection, placement, and performance management, and proposes a strategic HR framework — relevant to HRM systems design and organizational application

Adrias, A. F. (2022). Strategic Human Resource Management Framework for state Universities and Colleges in the Philippines. *International Journal of Scientific Research and Management (IJSRM)*, *10*(04), 3303–3311. https://doi.org/10.18535/ijsrm/v10i4.em8

Strategic Human Resource Management

*Figure 4.*

### *2.2 FOREIGN*

Provide at least three (3) FOREIGN related studies and projects/systems.

**Foreign Related Study 1.**

**"Global ERP/HRMS Software Implementation Case Study (USA)."**

A thesis analyzing success factors in a multinational HRMS rollout across sixteen countries. It examines cultural, communication, and management influences on HRMS implementation success.

House, D. (n.d.). *A Case Study of Success Factors Associated with a Global Implementation of ERP/HRMS Software*. DigitalCommons@UNO. https://digitalcommons.unomaha.edu/studentwork/1321/?utm_source

**A Case Study of Success Factors Associated with a Global Implementation of ERP/HRMS Software**

**Deanna House, MS**

**University of Nebraska, 2006**

**Advisor: Dr. Gert-Jan de Vreede**

This research observes a global implementation of Enterprise Resource Planning (ERP)/Human Resources Management System (HRMS) software at an International company in Omaha, Nebraska. The software was implemented in sixteen countries. Variables such as cultural differences, communication-distance, management support, trust, and fear of change were evaluated in the literature review. These variables have an impact on implementation success during global HRMS implementation. Further analyses on specific success factors faced with global implementations were evaluated using semi-structured interviews. The author prepared a questionnaire to further explore the data. Respondents rated questions related to management support the highest overall. An interesting find was that the semi-structured interview results indicated that the software chosen was not a perfect fit for the global community. The mean for Global HRMS Success was higher for respondents located in the United States than those located in other locations.

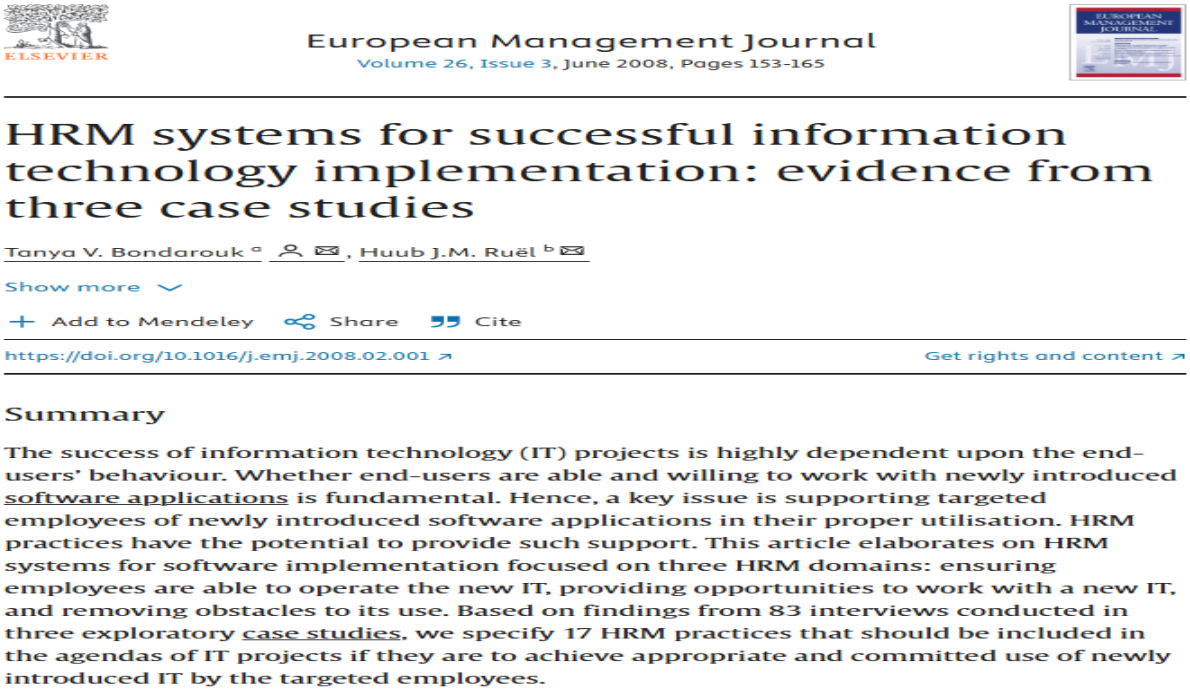
*Figure 5.*

**Foreign Related Study 2.**

**"HRM Systems & Organizational IT Implementation Case Studies."**

Research shows how HRM systems influence IT project success by motivating users, building skills, and supporting learning — crucial insights for HRMS adoption in organizations.

Tanya V. Bondarouk A, T. B., & J.M. Ruël B, J. M. (2008, June). HRM Systems for Successful Information Technology Implementation: Evidence from three case studies. Science Direct. https://linkinghub.elsevier.com/retrieve/pii/S0263237308000194



ELSEVIER

European Management Journal
Volume 26, Issue 3, June 2008, Pages 153-165

HRM systems for successful information technology implementation: evidence from three case studies

Tanya V. Bondarouk [a], Huub J.M. Ruël [b]

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https://doi.org/10.1016/j.emj.2008.02.001 Get rights and content

Summary

The success of information technology (IT) projects is highly dependent upon the end-users' behaviour. Whether end-users are able and willing to work with newly introduced software applications is fundamental. Hence, a key issue is supporting targeted employees of newly introduced software applications in their proper utilisation. HRM practices have the potential to provide such support. This article elaborates on HRM systems for software implementation focused on three HRM domains: ensuring employees are able to operate the new IT, providing opportunities to work with a new IT, and removing obstacles to its use. Based on findings from 83 interviews conducted in three exploratory case studies, we specify 17 HRM practices that should be included in the agendas of IT projects if they are to achieve appropriate and committed use of newly introduced IT by the targeted employees.

*Figure 6.*

**Foreign Related Study 3.**

**"HRMS Implementation Case Studies in Industry (SME Examples)."**

Collated case studies show HRMS deployments in various companies (transportation, manufacturing, service industries), with measurable impacts on HR efficiency, payroll accuracy, and administrative overhead.

*Client challenge*. (n.d.). https://www.scribd.com/document/956018460/Human-Resource-Management

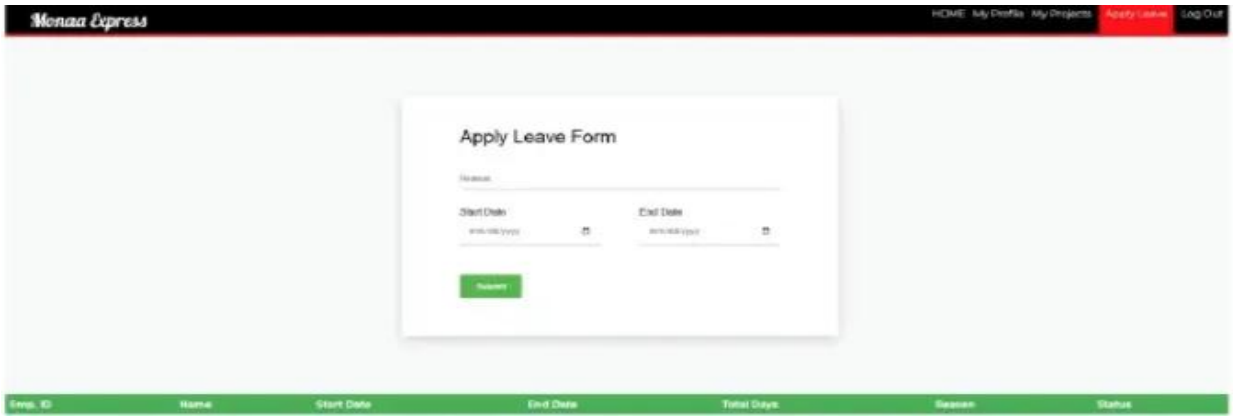



*Figure 7.*

*Figure 8.*

## 2.3 SYNTHESIS AND RELEVANCE TO THE STUDY (COMPARATIVE ANALYSIS)

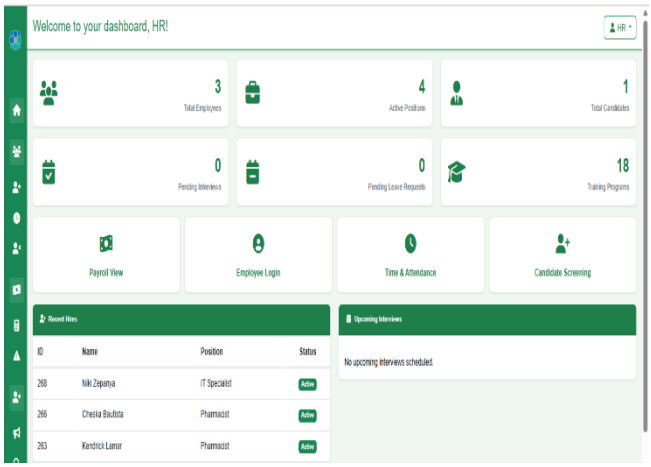
The following table summarizes the **Human Resource Management System (HRMS) modules**, comparing their current practices, digital implementation, HR framework alignment, global standards, and proposed features for **Healthway QualiMed Hospital, San Jose Del Monte**.

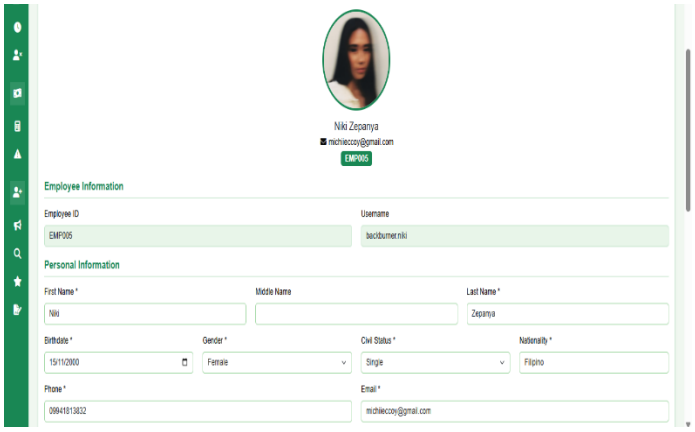
| HRMS Module / Feature | Ilocos Sur Polytechnic State College HRIS | Philippine Banking Sector e-HRM | State Universities HRM Framework | USA Multinational HRMS | HRM Systems & IT Adoption | SME Industry HRMS Case Studies | Your Project / System |
|---|---|---|---|---|---|---|---|
| Recruitment & Selection | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Onboarding | ✕ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Time & Attendance | ✕ | ✕ | ✕ | ✓ | ✓ | ✓ | ✓ |
| Termination / Offboarding | ✕ | ✕ | ✕ | ✓ | ✓ | ✓ | ✓ |
| Employee Management | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Payroll / Compensation | ✕ | ✕ | ✕ | ✓ | ✓ | ✓ | ✓ |
| System Integration / Accessibility | ✕ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

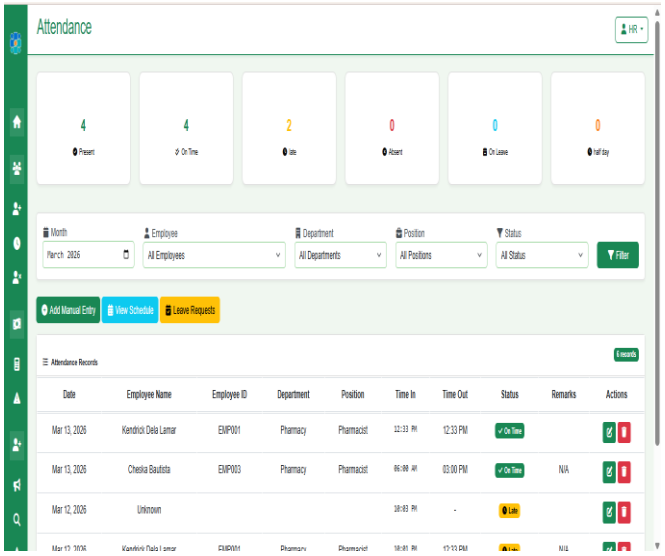
*Review of Related Studies and Projects/Systems Table no.1*

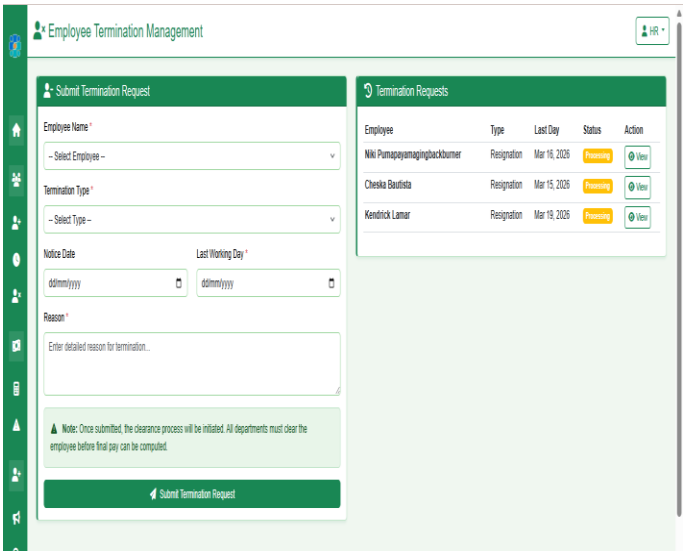
# CHAPTER 3 DESIGN/LAYOUT OF THE SYSTEM

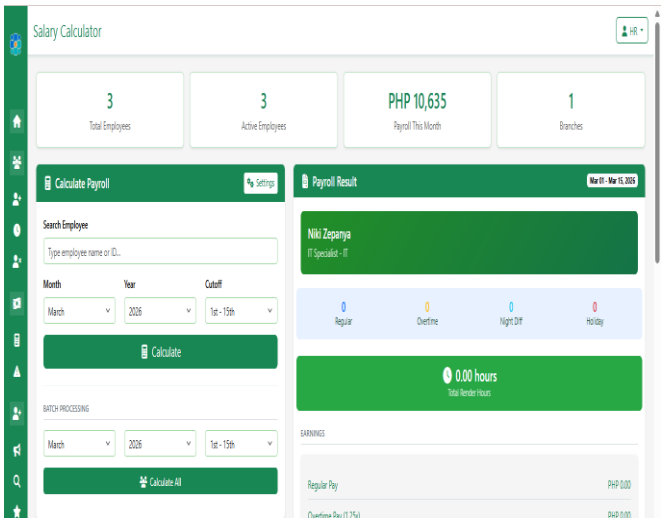
## 3.1 DESIGN/LAYOUT

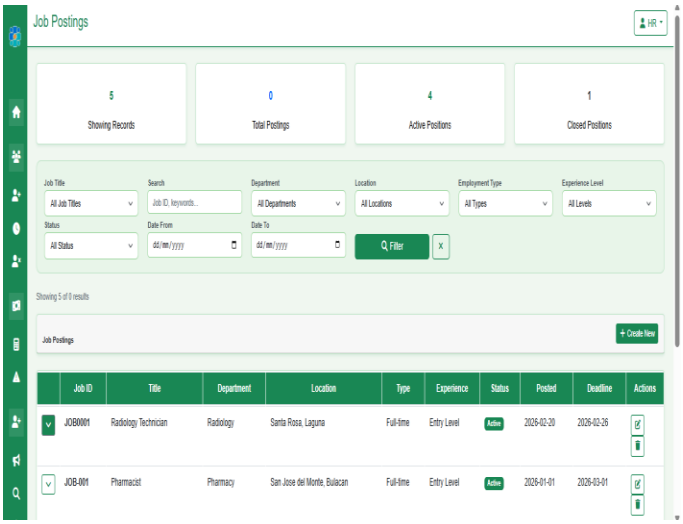


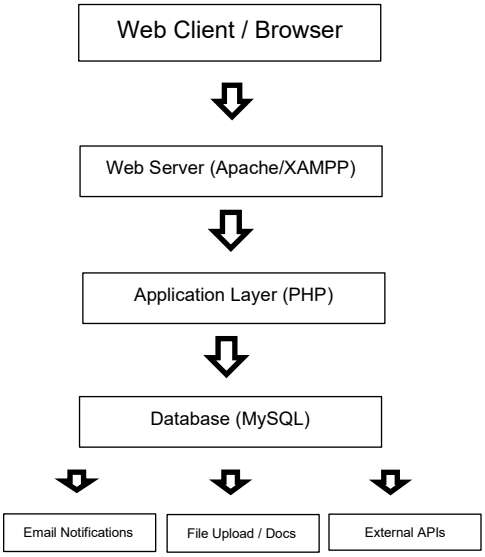





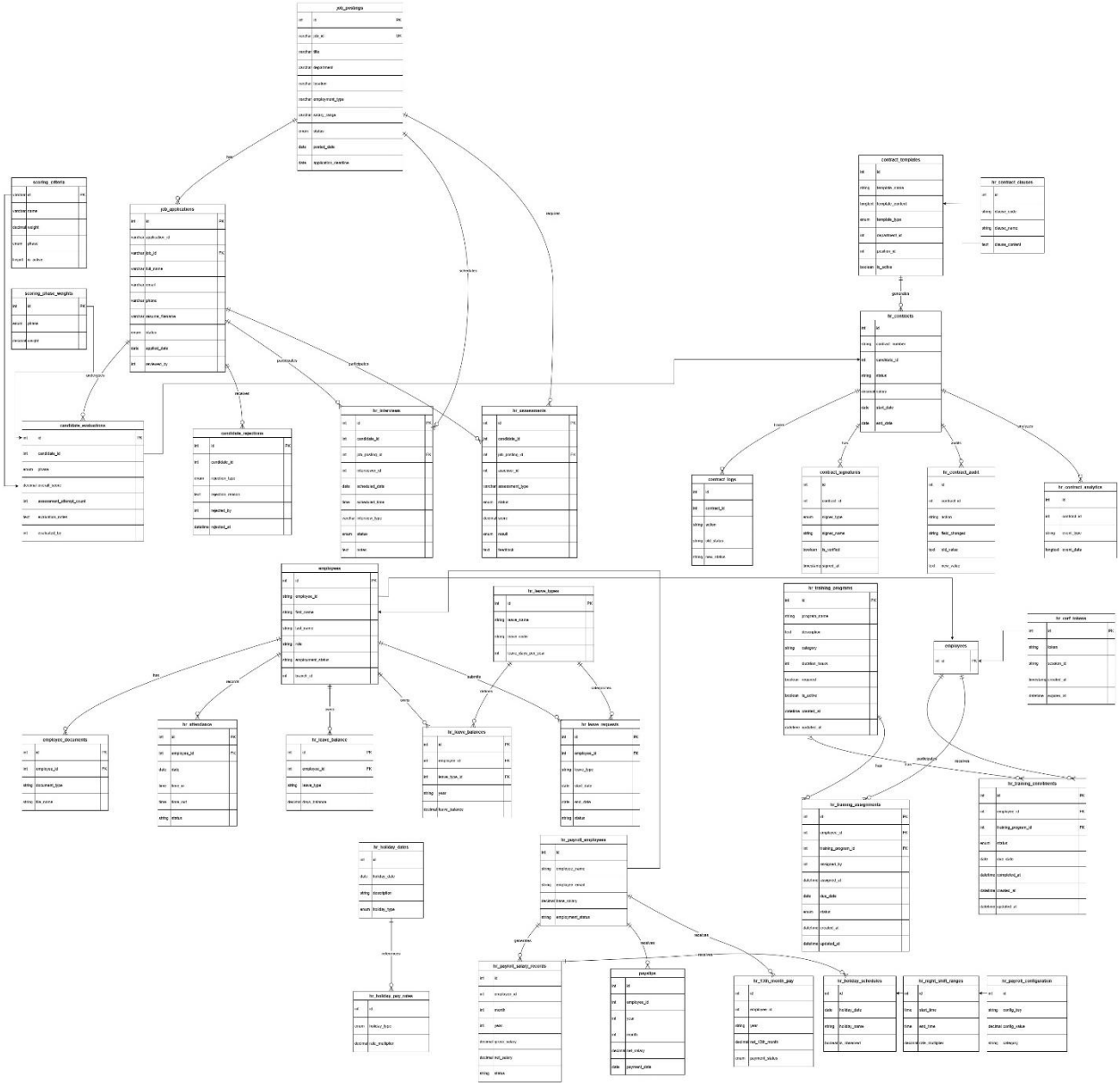




### 3.1.1 System Architecture



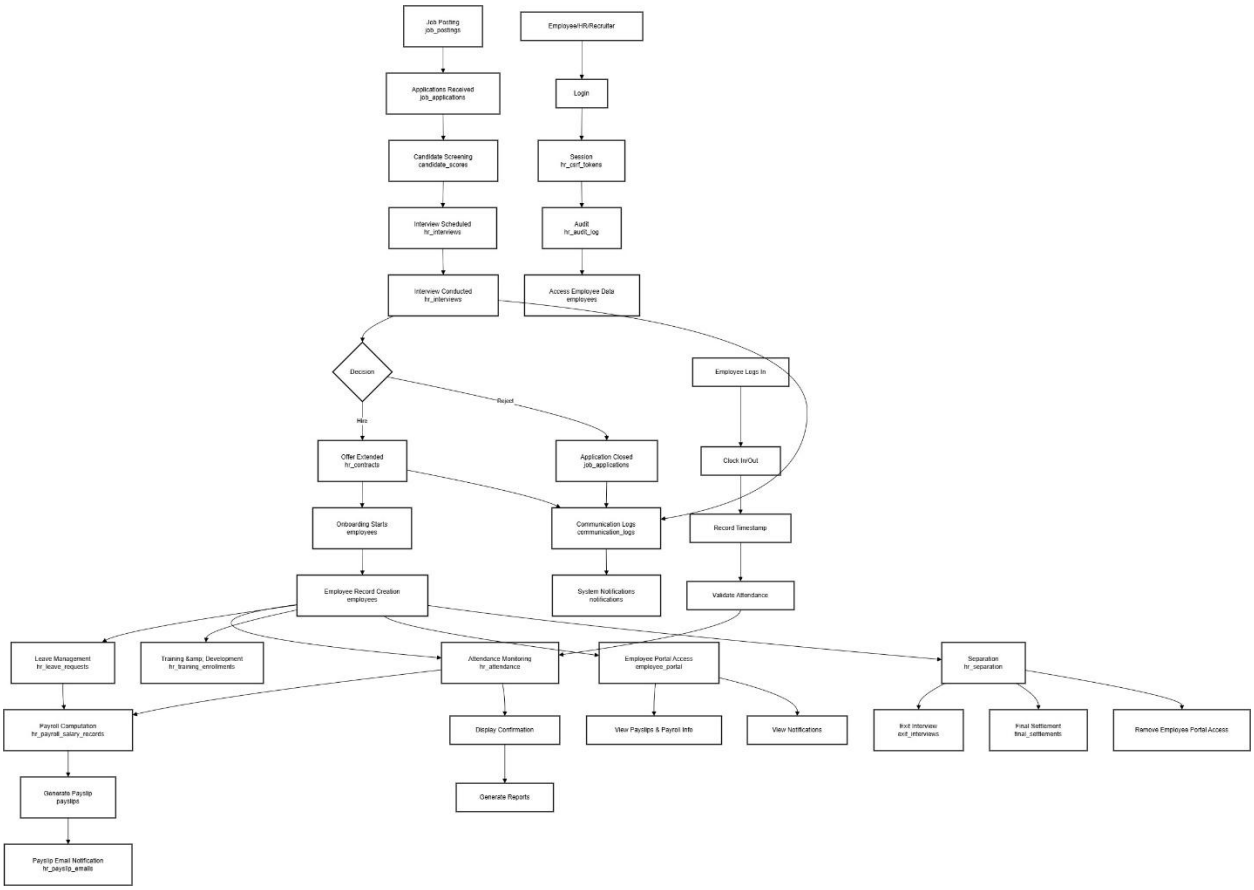

*System Architecture Figure No.1*

### 3.1.2 Business Process Architecture



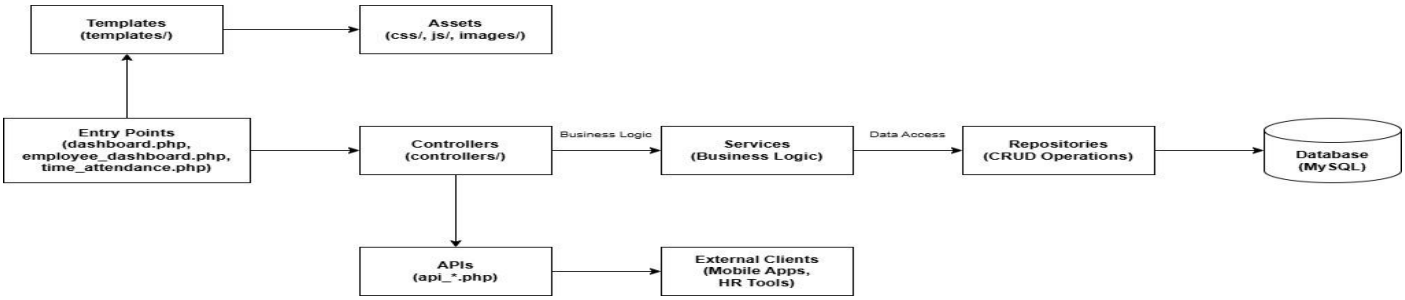
*Business Process Architecture Figure No. 1*

### 3.1.3 Flow Chart



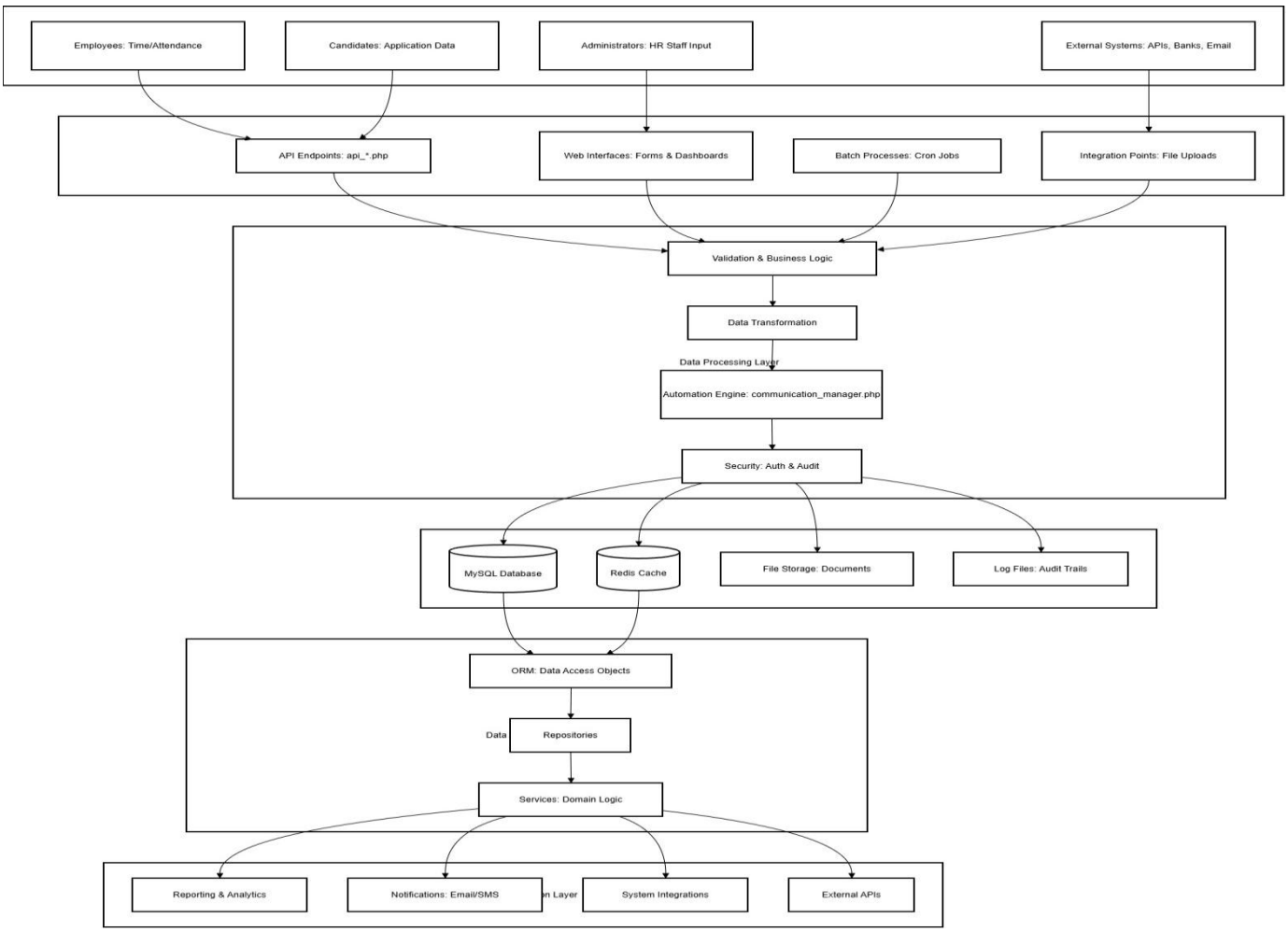

*Flow Chart Figure No. 1*

## 3.1.4 Application Architecture




*Application Architecture Figure No. 1*

## 3.1.5 Data Architecture



*Data Architecture Figure No. 1*

## 3.2 PROJECT/SYSTEM REQUIREMENTS

### System Requirements

- **Application Type:** Web-based HRMS supporting all HR modules: Recruitment, Selection, Onboarding, Time & Attendance, Employee Management, Payroll, Benefits, and Termination.
- **Database:** MySQL, PostgreSQL, or equivalent relational database to store employee, payroll, attendance, recruitment, and training data.
- **Security Measures:** Anti-virus, encryption for sensitive data, firewall protection, and role-based access control.
- **Reporting Tools:** Interactive dashboards, analytics, HR reports, and payroll summaries.
- **Optional Features:** Cloud integration for remote access, automatic updates, and multi-location access.

### 3.2.1 Policy and Regulations Requirements

1. **Data Privacy and Protection**
   The HRMS must comply with the **Data Privacy Act of 2012**, as it stores sensitive employee information including personal details, salary records, employment history, leave balances, training records, and benefits information.

- The system will implement **secure login**, **role-based access control**, and **data encryption** for both storage and transmission.
- Only authorized personnel (HR staff, administrators) can access confidential records.
- Employees can securely **view and update their personal information** via the self-service portal.

- Activity logs will record all updates for accountability and audit purposes.

2. **Compliance with Labor Laws**
   The system must follow the **Labor Code of the Philippines** to ensure employee rights and accurate compensation.

- Attendance, overtime, night, holiday, and leave calculations will be automated to ensure accuracy.
- Salary computation will comply with **minimum wage standards** and other statutory rules.
- Leave management and time-in/time-out features support proper monitoring and reporting.

3. **Government Contributions and Tax Requirements**
   The HRMS will automatically compute, deduct, and track mandatory contributions for:

- **SSS (Social Security System)**
- **PhilHealth**
- **Pag-IBIG Fund**
- **BIR (Income Tax)**
  Automated calculations and reporting ensure timely remittance and compliance with government regulations.

4. **Records Retention and Documentation**
   Employee, payroll, attendance, recruitment, onboarding, training, and termination records will be securely stored in the system for the legally required periods.

- Backup and recovery mechanisms will be in place to prevent data loss.
- The system supports audit compliance and organizational transparency.

5. **Access Control and Security**

- **Role-based access control** limits system access according to user roles: Administrator, HR Staff, and Employee.
- Strong password policies, automatic logout, and session management will prevent unauthorized access.
- All actions will be logged to maintain an **audit trail**.

6. **Equal Employment and Fair Treatment**
   The system enforces **objective and unbiased HR processes**: recruitment, evaluation, promotions, disciplinary actions, and termination.

- No discrimination based on gender, age, religion, ethnicity, or disability will be allowed.
- Candidate evaluation and hiring decisions are documented and auditable.

7. **Workplace Health, Safety, and Compliance**
   The system can record **workplace incidents, health-related leaves, and compliance with safety regulations**.

- Employee leave and attendance modules track absences related to health or safety concerns.
- Supports HR reporting for workplace safety compliance.

8. **Employee Portal Transparency**

- Employees can track **leave balances, payroll information, payslips, schedules, training progress, onboarding tasks, and benefits** securely.
- All portal activity is logged for accountability and data integrity.

### 3.2.2 Security Requirements

1. **Authentication**

- Users must log in via **unique username and password**.
- Password policies enforce **minimum length, character diversity**, and periodic changes.
- Only verified accounts gain system access.

2. **Role-Based Access Control (RBAC)**

- Administrators: full access to all modules and settings.
- HR Staff: manage employee records, payroll, benefits, attendance, recruitment, and termination.
- Employees: access only their personal data, leave requests, payslips, training, and onboarding tasks via the portal.
- Limits exposure of sensitive data.

3. **Data Protection and Encryption**

- Sensitive information (personal data, payroll, benefits) is **encrypted during storage and transmission**.
- Regular **system backups** ensure data recovery in case of failure, cyberattack, or accidental loss.

4. **Audit Trail and Monitoring**

- All user activity (logins, updates, deletions, payroll changes, leave approvals) is logged.
- Administrators can monitor suspicious or unauthorized activities.
- Supports compliance with the Data Privacy Act and internal HR policies.

5. **Secure Attendance and Payroll Management**

- Clock-in/out, leave submissions, and payroll calculations are securely logged.
- Automation prevents manipulation and ensures **accurate salary, overtime, and benefits calculations**.

6. **Employee Self-Service Security**

- Portal access uses **secure HTTPS connections**.
- Employees can only modify allowed fields (profile, leave requests, reimbursements).
- All transactions are tracked to prevent misuse.

### 3.2.3 Training Requirements

For the HR Management System (HRMS) to be effective, users must be properly trained to navigate and utilize all modules confidently. Without adequate training, even the most advanced system may not achieve its intended benefits.

**Training Plan:**

1. **HR Personnel and System Administrators:**

    - Pre-launch training sessions on managing employee records, processing payroll, monitoring attendance, approving leave, generating reports, and handling recruitment and onboarding workflows.
    - Administrators will also be trained on user account management, data security, backups, and system maintenance.

2. **Regular Employees (Self-Service Users):**

    - Basic orientation on using the employee portal for viewing personal data, submitting leave and reimbursement requests, checking payroll and payslips, monitoring onboarding tasks, and enrolling in training programs.
    - Simple demonstrations or step-by-step guides will be provided to make navigation easy.
2. **Ongoing Support:**
    - User manuals, quick reference guides, and FAQs will be available to support continuous learning and reduce reliance on technical support.
    - Refresher sessions may be conducted periodically when new features or updates are implemented.

### 3.2.4 Reporting Requirements

The HRMS will generate comprehensive reports to support decision-making, compliance, and workforce management.

1. **Employee Information Report:**

    - Complete profile of employees including personal information, job position, department, employment status, qualifications, and contact details.
2. **Attendance and Time Tracking Report:**
    - Records of attendance, working hours, overtime, absences, and tardiness to monitor staff punctuality and support payroll.
3. **Payroll Summary Report:**
    - Details of salaries, allowances, deductions, bonuses, and taxes for transparency and accurate compensation management.

4. **Leave Management Report:**
    - Tracks all types of leave (sick, vacation, maternity, emergency) and leave balances to ensure proper staffing levels.
5. **Recruitment and Hiring Report:**
    - Data on job postings, applications, shortlisted candidates, interview results, and newly hired employees to assess recruitment efficiency.
6. **Training and Development Report:**
    - Lists employee participation in training programs, certifications, and skills development activities to ensure workforce competencies are up to date.
7. **Department Staffing Report:**
    - Shows employee distribution across departments (e.g., emergency, nursing, laboratory, administration) to identify shortages or overstaffing.
8. **Employee Turnover Report:**
    - Tracks resignations, retirements, and terminations to analyze turnover rates and retention challenges.
9. **Compliance and Regulatory Report:**
    - Ensures licenses, certifications, mandatory trainings, and other regulatory requirements are current, supporting legal compliance and hospital accreditation.

### 3.3 Test Specification and Procedures

#### 3.3.1 Testing Objectives

- Ensure all HRMS modules function correctly and meet business requirements.
- Verify **data integrity** and security across all HR processes.
- Validate automation workflows, triggers, and scheduled jobs.
- Confirm system performance under normal and high-load conditions.

- Test integration with **external systems** such as email services and government APIs.

### 3.3.2 Test Types

- **Unit Testing**: Individual functions and methods in PHP classes and scripts.
- **Integration Testing**: Interactions and data flows between modules.
- **System Testing**: End-to-end workflows from recruitment to termination.
- **Performance Testing**: Load and stress testing with multiple concurrent users and large datasets.
- **Security Testing**: Authentication, authorization, access control, and data protection verification.
- **User Acceptance Testing (UAT)**: Business user validation of HR processes and system usability.

### 3.3.3 Test Environment Setup

- **Development Environment**: Local XAMPP/LAMP setup for unit and integration testing.
- **Staging Environment**: Mirror of production for system testing and UAT.
- **Test Data**: Anonymized employee, candidate, and payroll datasets for safe testing.
- **Test Tools**: PHPUnit for unit testing, Selenium for UI tests, JMeter for performance, and Postman for API testing.

### 3.3.4 Test Cases by Module

**Recruitment Module**

- Create, publish, and update job postings.
- Candidate application submission, validation, and tracking.
- Screening and shortlisting candidates.
- Interview scheduling and automated notifications.

- Assessment administration and scoring.
- Offer generation and contract creation.

**Onboarding Module**

- Employee profile creation from contract data.
- Document submission, verification, and approval.
- Enrollment in training programs.
- Automated welcome email generation.

**Time & Attendance Module**

- Clock-in/out functionality and timestamp validation.
- Attendance data integrity checks.
- Leave request submission and approval workflow.
- Overtime and holiday pay calculation accuracy.

**Payroll Module**

- Salary computation under various scenarios (regular, overtime, deductions, bonuses).
- Automatic calculation of government contributions (SSS, PhilHealth, Pag-IBIG) and taxes.
- Payslip generation and PDF export.
- Email distribution of payslips and payroll notifications.
- 13th month pay and final settlement calculations.

**Employee Management Module**

- Updating employee data and maintaining history.
- Benefits enrollment and reimbursement claims processing.
- Training progress and completion tracking.

**Termination Module**

- Resignation and termination workflow execution.
- Final pay and benefits settlement calculation.
- Exit interview scheduling.
- Archiving and retention of employee records for compliance.

### 3.3.5 Automation Testing Procedures

- Test **database triggers** for automatic data propagation between tables.
- Validate **email notifications** sent at correct workflow stages.
- Verify **scheduled jobs** (cron jobs) run correctly and on time.
- Test **API integrations** and data exchanges with external systems.

### 3.3.6 Performance Benchmarks

- Response time < 2 seconds for standard operations.
- Support for 100+ concurrent users.
- Payroll processing for 500 employees completed within 30 minutes.
- Email notifications delivered within 5 minutes of triggers.

### 3.3.7 Security Testing Procedures

- SQL injection and input validation tests.
- Cross-site scripting (XSS) vulnerability checks.
- Authentication bypass attempts.
- Data encryption verification (at rest and in transit).
- Role-based access control testing for all user levels.

### 3.3.8 Data Integrity Testing

- Validate foreign key constraints and relationships.
- Ensure consistency of data across related tables.
- Audit log completeness and accuracy checks.
- Backup and recovery procedures verification.

### 3.3.9 Test Execution and Reporting

- Track execution of test cases and results.
- Log, prioritize, and resolve bugs or errors.
- Perform regression testing after bug fixes or system updates.
- Generate **test summary reports** with pass/fail metrics.
- Obtain sign-off from stakeholders for production deployment.

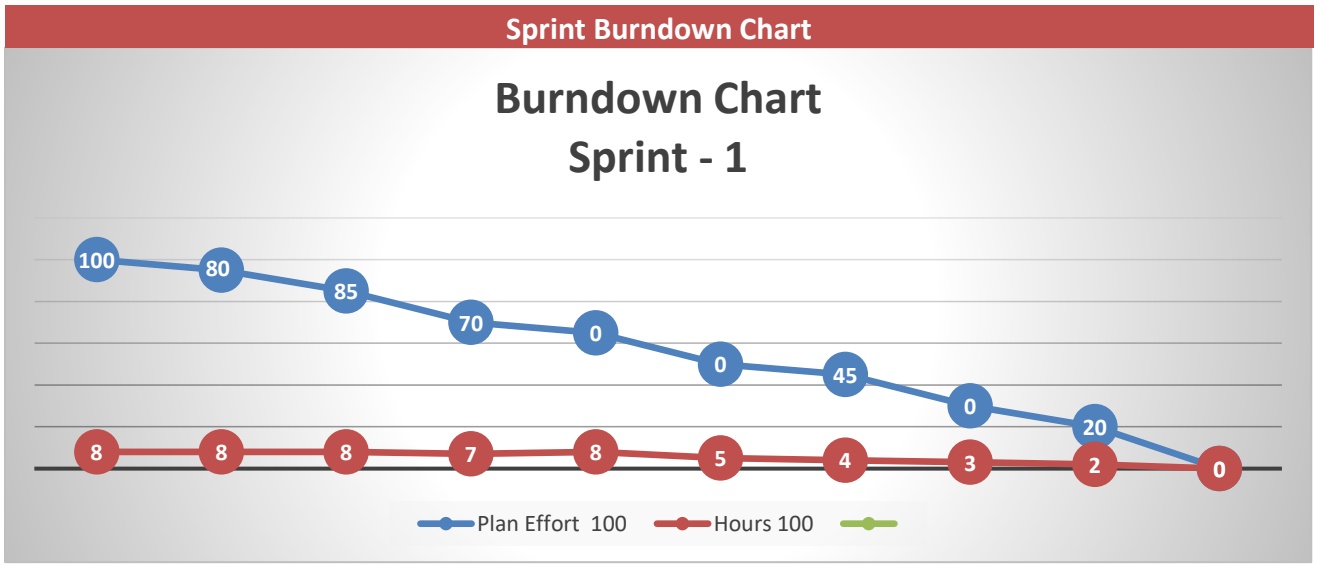
## 3.4 Sprint Planning

| Date Range | Scope | Status |
|---|---|---|
| Jan 18 – Jan 24 | Planning and requirements gathering for HRMS. Identified modules: Recruitment, Selection, Onboarding, Employee Management, Attendance & Time Tracking, Payroll, Benefits, Termination. | Finished |
| Jan 25 – Jan 31 | Design system architecture, database schema, and module interactions. Created initial UI for login, dashboards, employee portal, HR pages. | Finished |
| Feb 1 – Feb 7 | Development of Attendance & Time Tracking Module (employee login, clock-in/out, leave submission). | Finished |

| | | |
|---|---|---|
| Feb 8 – Feb 14 | Development of Leave Management and Approval Module (employee requests, HR/admin approval workflows). | Finished |
| Feb 15 – Feb 21 | Development of Payroll and Compensation Module (salary computation, deductions, benefits, payslip generation). | Finished |
| Feb 22 – Feb 28 | Implementation of Recruitment & Onboarding Modules (job posting, application tracking, candidate evaluation, employee profile creation, training enrollment). | Finished |
| Mar 1 – Mar 7 | Employee Management, Benefits & Reimbursement Module (performance tracking, benefits enrollment, claims management). | Finished |
| Mar 8 – Mar 14 | System testing, debugging, performance optimization, and UI/UX improvements across all modules. | In Progress |
| Mar 15 – Mar 18 | User Acceptance Testing (UAT), final documentation, and preparation for production deployment. | Planned |

*Sprint Planning Table no. 1*

## 3.4 Sprint Timeline




*Sprint Burndown Chart Figure no.1*

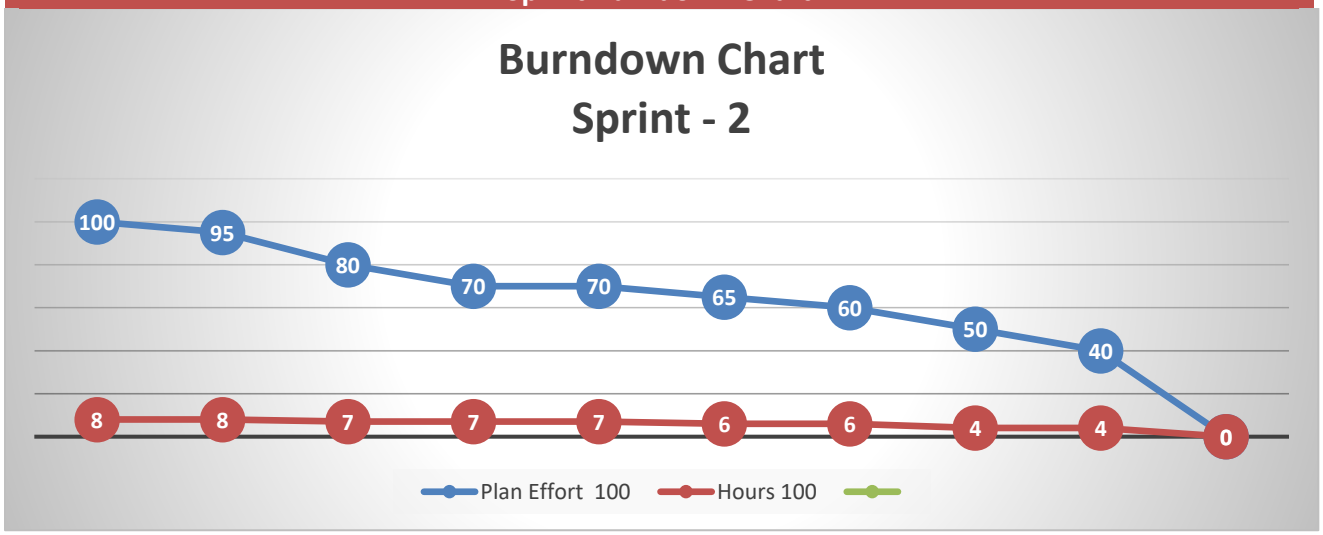
| Sprint No.1 Backlog | | | Sprint No. | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Product Backlog Item ID | User Stories (Features) | Initial Estimate | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 | Day 6 | Day 7 | Day 8 | Day 9 | Day 10 | Day 11 | Day 12 | Day 13 | Day14 |
| 001 | Documentation - Chapter 1 | 2 days | X | X | | | | | | | | | | | | |
| 002 | Documentation - Chapter 2 | 2 days | | | X | X | | | | | | | | | | |
| 003 | Documentation - Chapter 3 | 2 days | | | | X | X | | | | | | | | | |
| 004 | Data Gathering | 2 weeks | | | | | | X | X | X | X | X | X | X | X | X |
| 005 | Planning | 8 hours | X | | | | | | | | | | | | | |
| 006 | Monitoring | ongoing | X | X | X | X | X | X | X | X | X | X | X | X | X | X |
| | **Total Effort** | 2 weeks | | | | | | | | | | | | | | |

*Sprint 1 Backlog chart Table No.1*

# BESTLINK COLLEGE OF THE PHILIPPINES

IPO Road, Barangay Minuyan Proper City of San Jose Del Monte, Bulacan

**Sprint Burndown Chart**



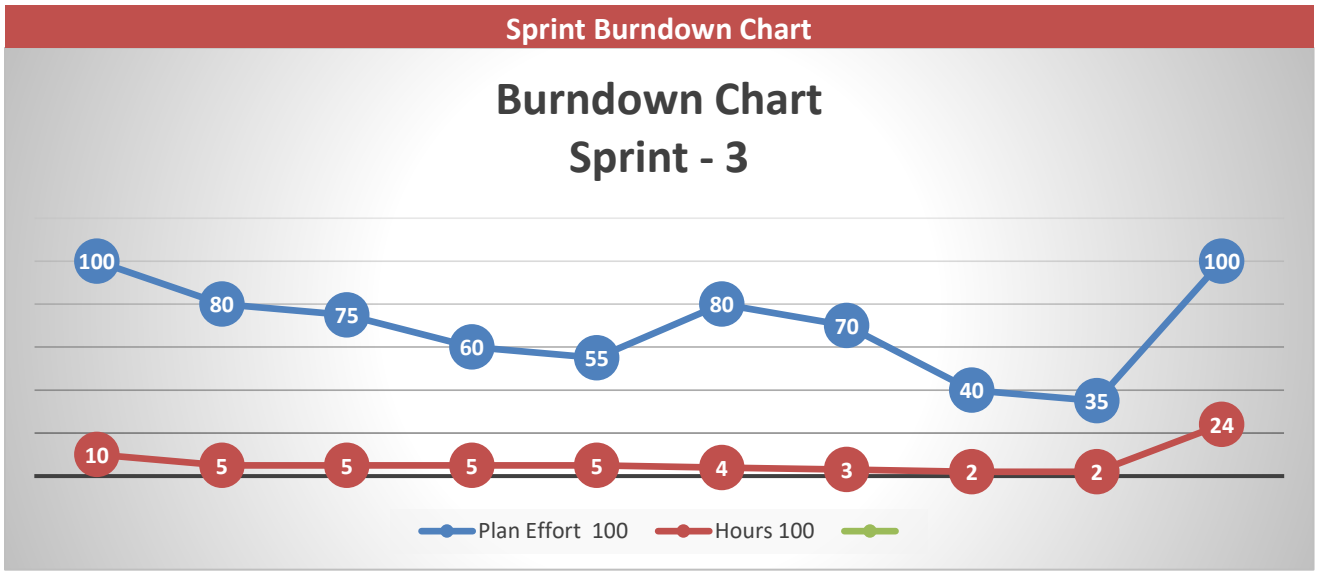

*Sprint Burndown Chart Figure no.2*

| Sprint No.2 Backlog | | | Sprint No. 2 | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Product Backlog Item ID | User Stories (Features) | Initial Estimate | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 | Day 6 | Day 7 | Day 8 | Day 9 | Day 10 | Day 11 | Day 12 | Day 13 | Day14 |
| 201 | Job Posting Management | 8hrs | X | X | | | | | | | | | | | | |
| 202 | Recruitment Management | 8hrs | | | X | X | | | | | | | | | | |
| 203 | Selection Management | 8hrs | | | | | X | X | | | | | | | | |
| 204 | Contract Generation | 8hrs | | | | | | | X | X | | | | | | |
| 205 | Onboarding Management | 8hrs | | | | | | | | | X | X | | | | |
| 206 | Employee Management | 16hrs | | | | | | | | | | | X | X | X | X |
| | Total Effort | 2 weeks | | | | | | | | | | | | | | |

***Sprint 2 Backlog chart Table No. 2***

# BESTLINK COLLEGE OF THE PHILIPPINES

IPO Road, Barangay Minuyan Proper City of San Jose Del Monte, Bulacan




*Sprint Burndown Chart Figure no.3*

| Sprint No.3 Backlog | | | Sprint No. 3 | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Product Backlog Item ID | User Stories (Features) | Initial Estimate | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 | Day 6 | Day 7 | Day 8 | Day 9 | Day 10 | Day 11 | Day 12 | Day 13 | Day14 |
| 207 | Employee Self-Service Portal | 8 hrs | X | X | | | | | | | | | | | | |
| 208 | Attendance Management | 8 hrs | | | X | X | | | | | | | | | | |
| 209 | Time & Attendance (Clock-In) | 8 hrs | | | | X | X | | | | | | | | | |
| 210 | Payroll and Compensatio n | 16 hrs | | | | | | X | X | | | | | | | |
| 211 | Benefits and Reimbursem ents | 8 hrs | | | | | | | | X | X | | | | | |
| 212 | Termination Management | 8 hrs | | | | | | | | | X | X | | | | |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 213 | HR Dashboard Reporting | 8 hrs | | | | | | | | | | | X | X | | |
| 214 | Final Review & Integration | 24 hrs | | | | | | | | | | | X | X | X | X |
| | Total Effort | 2 weeks | | | | | | | | | | | | | | |

***Sprint 3 Backlog chart Table No. 3***

## 3.6 Assumptions and Constraints

### Assumptions

The development of the system is based on several assumptions that are considered to be true throughout the implementation of the project.

• **Availability of Computer Resources**
It is assumed that the hospital has sufficient computer equipment and a stable internet connection required for operating the system efficiently.

• **User Cooperation and System Adoption**
The study assumes that hospital management and HR personnel are willing to utilize the system and adopt the digital processes introduced by the system.

• **Accuracy of Data Entered in the System**
It is assumed that all information entered into the system will be accurate, valid, and regularly updated by authorized personnel to ensure reliable records.

• **User Training and Basic System Knowledge**
The study assumes that users will receive proper orientation or training to understand the basic functions and operations of the system.

• **Authorized Access to the System**
It is assumed that only authorized users will be granted access to specific system features to ensure the confidentiality and protection of employee information.

**Constraints**

Constraints refer to the limitations that may affect the development, implementation, or performance of the system.

• **Budget Limitations**
The hospital may have limited financial resources for the development, deployment, and maintenance of the system.

• **User Familiarity with Technology**
Some hospital staff may have limited experience with computer-based systems, which may require additional time and training for them to adapt to the new system.

• **Dependence on Existing Technology Infrastructure**
The system's performance may depend on the availability and reliability of the hospital's existing hardware, software, and network infrastructure.

• **Data Security and Privacy Requirements**
The system must ensure the protection of sensitive employee information by implementing appropriate data security measures and privacy controls.

• **Limited System Coverage**
The system primarily focuses on Human Resource Management functions and may not fully integrate with other hospital systems unless additional development and integration are performed.

### 3.8 Sprint Backlog Output

| Task Number | Task Name | Output | Test Result | Accepted by Stake Owner |
|---|---|---|---|---|
| 1 | Job Posting Management | Job posting module for creating and managing job openings | Passed | HR Manager |
| 2 | Recruitment Management | Recruitment module for tracking applicants | Passed | HR Manager |
| 3 | Selection Management | Candidate screening and selection feature | Passed | HR Manager |
| 4 | Contract Generation | Automated contract generation for hired applicants | Passed | HR Manager |
| 5 | Onboarding Management | Employee onboarding and orientation tracking module | Passed | HR Manager |
| 6 | Employee Management | Employee information management system | Passed | HR Manager |
| 7 | Employee Self-Service Portal | Portal for employees to view personal records and requests | Passed | HR Manager |
| 8 | Attendance Management | Attendance monitoring and record management | Passed | HR Manager |
| 9 | Time and Attendance (Clock-In) | Employee clock-in and time tracking system | Passed | HR Manager |
| 10 | Payroll and Compensation | Payroll calculation and salary management module | Passed | HR Manager |
| 11 | Benefits and Reimbursements | Employee benefits and reimbursement tracking | Passed | HR Manager |
| 12 | Termination Management | Employee termination processing and record update | Passed | HR Manager |
| 13 | HR Dashboard Reporting | HR dashboard with reports and analytics | Passed | HR Manager |
| 14 | Final Review and Integration | Integrated system with all HR modules functioning | Passed | System Administrator |

*Sprint Backlog Output Table no. 1*

## 3.8 Sprint Product Road Map

| | | | | | | |
|---|---|---|---|---|---|---|
| | | Scrum Master | | Marycris Eliaga | | |
| | | Programmer | | Maria Cheska Jalotjot | | |
| | | Document Specialist | | Jade Clarenz Lorenzo and Cedrick Secosana | | |
| | | Business Specialist | | Jonha Mhae Azucena | | |
| | | System Analyst | | Phil Vincent Lope | | |
| **Scope** | **Task name** | **Responsible** | **Duration** | **January** | **February** | **March** |
| Documentation | Data Gathering | Business Specialist | 2 weeks | ✓ | | |
| Documentation | Chapter 1–3 Preparation | Document Specialist | 2 weeks | ✓ | | |
| Planning | Requirements Analysis | System Analyst | 1 week | ✓ | | |
| Planning | System Design | System Analyst | 1 week | ✓ | | |
| Development | Job Posting Management | Programmer | 1 week | | ✓ | |
| Development | Recruitment Management | Programmer | 1 week | | ✓ | |
| Development | Selection Management | Programmer | 1 week | | ✓ | |
| Development | Contract Generation | Programmer | 1 week | | ✓ | |
| Development | Onboarding Management | Programmer | 1 week | | ✓ | |
| Development | Employee Management Module | Programmer | 2 weeks | | ✓ | |
| Development | Attendance Management | Programmer | 1 week | | ✓ | |
| Development | Time & Attendance (Clock-In) | Programmer | 1 week | | ✓ | |
| Development | Payroll and Compensation | Programmer | 2 weeks | | ✓ | |
| Development | Benefits and Reimbursements | Programmer | 1 week | | ✓ | |
| Development | Termination Management | Programmer | 1 week | | ✓ | |
| Development | HR Dashboard Reporting | Programmer | 1 week | | ✓ | |
| Integration | Final Review & Integration | Programmer & System Analyst | 1 week | | | ✓ |
| Testing | System Testing and Debugging | Scrum Master & Programmer | 1 week | | | ✓ |
| Documentation | Final Documentation & Revision | Document Specialist | 1 week | | | ✓ |

*Sprint Product Road Map Table no. 1*